16.12.25

Lecture 19

PLEASE EVALUATE THE COURSE! →

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Assumption (Approximation property): Let $(V_h)_{h\in(0,1)} \subset V$ be a sequence of Galerkin finite element spaces that for some $r \in \mathbb{N}$ and constant C satisfies

$$\inf_{v_h \in V_h} \|v - v_h\|_{L^2(\Omega)} \leq C h^r \|v\|_{H^r(\Omega)}$$

and

$$\inf_{v_h \in V_h} \|v - v_h\|_{H^1(\Omega)} \leq C h^{r-1} \|v\|_{H^r(\Omega)}$$

for all $v \in H^r(\Omega)$. Note that C does not depend on $v \in H^r(\Omega)$!

Important: in particular holds true for piecewise linear elements!

Theorem (Lemma of Céa). Let the assumptions (Lax-Milgram + approximation property) be satisfied with $r = 2$. Assume that $u \in H^2(\Omega)$ is the exact solution of (VP) and let $u_h \in V_h$, $h \in (0,1)$, be a sequence of FEM approximations. Then

$$\|u_h - u\|_{H^1(\Omega)} \leq \tilde{C} h \|u\|_{H^2(\Omega)} \quad \text{(more generally: } \|u-u_h\|_{H^1(\Omega)} \leq C h^{r-1} \|u\|_{H^r(\Omega)}\text{)}$$

Holds true similarly for the L^2-norm.

Theorem Let the assumptions (Lax-Milgram + approximation property) be satisfied with $r = 2$. Assume that $u \in H^2(\Omega)$ is the exact solution of (VP) and let $u_h \in V_h$, $h \in (0,1)$, be a sequence of FEM approximations. Then

$$\|u - u_h\|_{L^2(\Omega)} \leq C h^2 \|u\|_{H^2(\Omega)}.$$

Comparison: FDM vs FEM analysis

- FDM required $u \in C^4(\bar{\Omega})$ for convergence order 2 w.r.t. $\|\cdot\|_\infty$.
- FEM required $u \in H^2(\Omega)$ for convergence order 2 w.r.t. $\|\cdot\|_{L^2(\Omega)}$.
- Higher order Lagrange elements can improve convergence order IF exact solution is more regular, e.g., $u \in H^3(\Omega)$.

FEM for parabolic problems: main idea

Given $\Omega \subset \mathbb{R}^d$ with polygonal boundary, source term $f: [0,T] \times \Omega \to \mathbb{R}$ and $u_0: \Omega \to \mathbb{R}$ let us consider

$$
\begin{aligned}
u_t - \Delta u &= f && \text{in } (0,T) \times \Omega \\
u(t,x) &= 0 && \text{for all } t \in (0,T),\ x \in \partial\Omega \\
u(0,x) &= u_0(x) && \text{for all } x \in \Omega
\end{aligned}
$$

Similar as before, let us multiply with a test function $v \in H_0^1(\Omega) = V$ (time independent) and integrate over Ω ...

$$\int_\Omega u_t(t,x)\, v(x)\, dx + a(u(t,\cdot), v) = \int_\Omega f(t,x)\, v(x)\, dx$$

We obtain the following <u>time-dependent</u> variational problem

Find $u: [0,T] \times \Omega \to \mathbb{R}$ such that

a) For every $t \in (0,T)$ we have $u(t,\cdot) \in H_0^1(\Omega)$.

b) $\langle u_t(t,\cdot), v \rangle_{L^2(\Omega)} + a(u(t,\cdot), v) = \langle f(t,\cdot), v \rangle_{L^2(\Omega)}$ for all $v \in H_0^1(\Omega)$.

c) $u(0,x) = u_0(x)$ for all $x \in \Omega$.

We start with a spatial (semi-)discretization: consider $(V_h)_{h \in (0,1)}$ a sequence of finite element spaces with refinement parameter $h \in (0,1)$.

For example: $V_h = \operatorname{span}\{\varphi_j \mid j = 1, \dots, N_h\}$, φ_j hat/pyramid fcts

$\to$ we obtain a semi-discrete problem: find $u_h: [0,T] \to V_h$ s.t.

$$\frac{d}{dt} \langle u_h(t), v_h \rangle_{L^2(\Omega)} + a(u_h(t), v_h) = \langle f(t), v_h \rangle_{L^2(\Omega)}$$

for all $t \in (0,T)$, $v_h \in V_h$

$$u_h(0) = u_{0,h} \in V_h$$

We will assume $u_h(t)$ to be of the form

$$\left.
\begin{aligned}
u_h(t) &= \sum_{j=1}^{N_h} \alpha_j(t)\, \varphi_j \in V_h \\
u_h(0) &= u_{0,h} = \sum_{j=1}^{N_h} \alpha_j(0)\, \varphi_j
\end{aligned}
\right\} \text{here, } \alpha_j : [0,T] \to \mathbb{R} \text{ to be determined}$$

The semi-discrete problem with $v_h = \varphi_\ell$ now yields

$$\frac{d}{dt} \langle u_h(t), \varphi_\ell \rangle_{L^2(\Omega)} + a(u_h(t), \varphi_\ell) = \langle f(t), \varphi_\ell \rangle_{L^2(\Omega)}$$

$$\Leftrightarrow \sum_{j=1}^{N_h} \dot{\alpha}_j(t) \langle \varphi_j, \varphi_\ell \rangle_{L^2(\Omega)} + \sum_{j=1}^{N_h} \alpha_j(t)\, a(\varphi_j, \varphi_\ell) \overset{!}{=} \langle f(t), \varphi_\ell \rangle_{L^2(\Omega)}, \quad \ell = 1, \dots, N_h$$

Matrix-vector form: $M_h \dot{\alpha}(t) + A_h \alpha(t) = f_h(t)$

where $\alpha(t) = [\alpha_1(t), \dots, \alpha_{N_h}(t)]^T$, $M_h = [\langle \varphi_j, \varphi_\ell \rangle_{L^2(\Omega)}]_{j,\ell=1}^{N_h}$ "mass matrix"

$A_h = [a(\varphi_j, \varphi_\ell)]_{j,\ell=1}^{N_h}$ "stiffness matrix"

$f_h(t) = [\langle f(t), \varphi_\ell \rangle_{L^2(\Omega)}]_{\ell=1}^{N_h}$ "load vector"

$\Rightarrow$ from here, we use a method for the numerical solution/approx. of (linear) ordinary differential equations